Samuel Baumgartner

7th-semester BSc Electrical Engineering, ETH Zurich

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Profile

Electrical Engineering student with a GPA of 5.7/6.0 (top 5%), planning an MSc in Robotics. Combines analytical training with leadership, teaching, international study, and rapid project work, with an interest in public systems, sustainability, and entrepreneurship.

Education

ETH Zurich *BSc Electrical Engineering and Information Technology* *2023–present*

- 141 ECTS completed; focus on control, signal processing, embedded systems, and robotics.
- Selected coursework: Control Systems, Computational Thinking, and Signal and System Theory.
- Broader-context coursework: Swiss Environmental Policy; Soccer Analytics.

Institute of Science Tokyo, Japan *Exchange semester in Tokyo* *Apr–Aug 2026*

- Interdisciplinary coursework including data analytics and machine learning, biomechanics and digital human modelling, optics, and solid dynamics.
- Adapted to a different academic culture while studying Japanese at approximately JLPT N4 level.

Leadership and Experience

ETH Zurich *Teaching Assistant* *2024–present*

- Lead weekly exercise sessions and support students in circuit analysis and magnetic fields.
- Explain complex material to students with different levels of preparation and provide individual feedback.

Swiss Armed Forces *Sergeant, technical function* *2022–2023*

- Led a 25-person unit and held technical responsibility for digital systems in a mission-critical environment.
- Developed discipline, clear communication, and decision-making under pressure.

Selected Projects

UrbanLens *NASA Space Apps Challenge, 3rd place* *Oct 2025*

- Built the Python API, analysis pipeline, and public-transport accessibility layer for a healthy-cities platform combining mobility, heat, pollution, vegetation, population, and land-use data.
- Worked in a five-person team to turn heterogeneous evidence into an interactive tool for comparing urban priorities.

Flora AI *StartHack, 2nd of 26 teams* *Mar 2026*

- Built an agent-driven indoor-farming and nutrition simulation combining AI, environmental sensors, embedded hardware, and a web interface in 36 hours.

Guidely *Swiss AI Hack, 2nd of 40 teams* *Nov 2025*

- Developed the core AI guidance pipeline and desktop overlay within 23 hours while teammates focused on user value, business model, and pitch.

ID Scan *Young Talents Hackathon, 1st place* *Oct 2024*

- Built an ID scanning app for Zurich Airport that reads identity documents and fills in the details automatically.

Skills and Languages

Methods and tools: Python, C++, data analysis, machine learning, control systems, embedded systems, full-stack prototyping, Git, Linux, Docker.

Languages: German (native), English (fluent), French (intermediate, Swiss Matura), Japanese (approximately JLPT N4, self-assessed).

Interests: Robotics, technology policy, sustainable systems, entrepreneurship, beekeeping, and fitness.